Practice Set A

Class 12 Mathematics, Chapter 6: Application of Derivatives

Maximum marks: **52** **20** questions

Name _____ Date _____ Score _____ / 52

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ, Board-pattern — 1 mark

The radius of a circular plate is increasing at the rate of 0.2 cm/s. The rate at which its area is increasing when the radius is 5 cm is:

(A) π cm^2/s **(B)** 2π cm^2/s **(C)** 10π cm^2/s **(D)** 25π cm^2/s

2 MCQ, CBSE SQP 2024-25 — 1 mark

The largest interval in which the function f defined by $f(x) = e^x$ is strictly increasing, is:

(A) $[1, \infty)$ **(B)** $(-\infty, 0)$ **(C)** $(-\infty, \infty)$ **(D)** $(0, \infty)$

3 MCQ, CBSE SQP 2025-26 — 1 mark

A function $f(x) = 10 - x - 2x^2$ is increasing on the interval:

(A) $\left(-\infty, -\frac{1}{4}\right]$ **(B)** $\left(-\infty, \frac{1}{4}\right)$ **(C)** $\left[-\frac{1}{4}, \infty\right)$ **(D)** $\left[-\frac{1}{4}, \frac{1}{4}\right]$

4 MCQ, CBSE 2025 — 1 mark

The absolute maximum value of the function $f(x) = x^3 - 3x + 2$ in $[0, 2]$ is:

(A) 0 **(B)** 2 **(C)** 4 **(D)** 5

5 MCQ, Board-pattern — 1 mark

The local minimum value of $f(x) = x^2 + \frac{16}{x}$, $x > 0$, is:

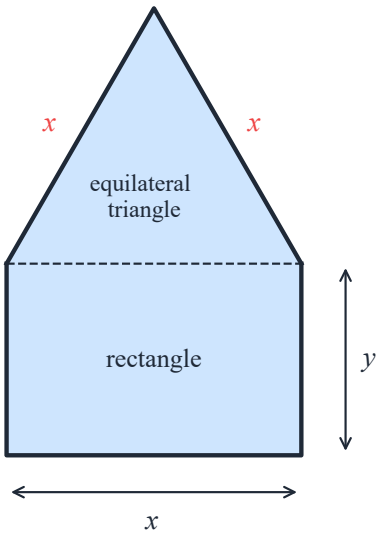
(A) 2 **(B)** 8 **(C)** 12 **(D)** 16

Q No.	Question	Marks
6	Assertion-Reason, CBSE SQP 2023-24 Let $f(x)$ be a polynomial function of degree 6 such that $\frac{d}{dx}\big(f(x)\big) = (x-1)^3(x-3)^2$. **Assertion (A):** $f(x)$ has a minimum at $x = 1$. **Reason (R):** When $\frac{d}{dx}\big(f(x)\big) < 0$ for all $x \in (a-h, a)$ and $\frac{d}{dx}\big(f(x)\big) > 0$ for all $x \in (a, a+h)$, where h is an infinitesimally small positive quantity, then $f(x)$ has a minimum at $x = a$, provided $f(x)$ is continuous at $x = a$. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer, CBSE 2025 The side of an equilateral triangle is increasing at the rate of 3 cm/s. At what rate is its area increasing when the side of the triangle is 15 cm?	2 marks

Level 2

19 marks

8	Short answer, CBSE 2025 Find the intervals in which the function $f(x) = 5x^3 - 3x^2$ is (i) increasing (ii) decreasing.	3 marks
9	Short answer, CBSE SQP 2025-26 A spherical ball of ice melts in such a way that the rate at which its volume decreases at any instant is directly proportional to its surface area. Prove that the radius of the ice ball decreases at a constant rate.	2 marks
10	Short answer, PYQ-style Find the local maximum and local minimum values of $f(x) = 3x^4 + 4x^3 - 12x^2 + 12$.	3 marks
11	Short answer, CBSE SQP 2023-24 Find the interval/s in which the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = xe^x$ is increasing.	3 marks
12	Short answer, CBSE SQP 2024-25 A kite is flying at a height of 3 metres and 5 metres of string is out. If the kite is moving away horizontally at the rate of 200 cm/s, find the rate at which the string is being released.	3 marks
13	Short answer, CBSE 2025 Find the absolute maximum and absolute minimum of the function $f(x) = 2x^3 - 15x^2 + 36x + 1$ on $[1, 5]$.	3 marks
14	Short answer, CBSE SQP 2023-24 Find the maximum profit that a company can make, if the profit function is given by $P(x) = 72 + 42x - x^2$, where x is the number of units and P is the profit in rupees.	2 marks

Q No.	Question	Marks
Level 3		**25 marks**
15	Long answer, Board-pattern Show that among all right-angled triangles with a given hypotenuse h, the isosceles triangle has the maximum area. Also find this maximum area.	5 marks
16	Long answer, PYQ-style Show that the altitude of the right circular cone of maximum volume that can be inscribed in a sphere of radius r is $\frac{4r}{3}$. Also show that the maximum volume of the cone is $\frac{8}{27}$ of the volume of the sphere.	5 marks
17	Long answer, PYQ-style A window is in the form of a rectangle surmounted by an equilateral triangle. The perimeter of the window is 12 m. Find the dimensions of the rectangle that will produce the largest area of the window. 	5 marks
18	Short answer, PYQ-style Find the intervals in which the function $f(x) = \sin^4 x + \cos^4 x$, $0 \le x \le \frac{\pi}{2}$, is strictly increasing or strictly decreasing.	3 marks
19	Short answer, PYQ-style A water tank has the shape of an inverted right circular cone with its axis vertical and vertex lowermost. Its semi-vertical angle is $\tan^{-1}(0.5)$. Water is poured into it at a constant rate of $5\ \mathrm{m}^3/\mathrm{h}$. Find the rate at which the level of the water is rising at the instant when the depth of water in the tank is 4 m.	3 marks

Q No.	Question	Marks
20	Case study, CBSE SQP 2025-26 **Case Study:** LED bulbs are energy-efficient because they use much less electricity than traditional bulbs while producing the same amount of light. A company manufactures a new type of energy-efficient LED bulb. The cost of production and the revenue generated by selling x bulbs (in an hour) are modelled as $C(x) = 0.5x^2 - 10x + 150$ and $R(x) = -0.3x^2 + 20x$ respectively, where $C(x)$ and $R(x)$ are both in ₹. Use the given models to answer the following questions. **(i)** Derive the profit function $P(x)$. — 1 mark **(ii)** Find the critical points of $P(x)$. — 1 mark **(iii)(a)** Determine whether the critical points correspond to a maximum or a minimum profit by using the second derivative test. — 2 marks OR **(iii)(b)** Identify the possible practical value of x (i.e. the number of bulbs that can realistically be produced and sold) that can maximise the profit, if the resources available and the expenditure on machines allow the company to produce a minimum of 10 but not more than 18 bulbs per hour. Also calculate the maximum profit. — 2 marks	4 marks

Answer key

Chapter 6: Application of Derivatives, Practice Set A. Full step-wise solutions are in the solutions sheet.

Level 1

1	(B) $2\pi \text{ cm}^2/\text{s}$
2	(C) $(-\infty, \infty)$
3	(A) $\left(-\infty, -\frac{1}{4}\right]$
4	(C) 4
5	(C) 12
6	(A) Both true; R is the correct explanation of A.
7	Area increases at $\frac{45\sqrt{3}}{2} \text{ cm}^2/\text{s}$

Level 2

8	(i) Increasing in $(-\infty, 0] \cup \left[\frac{2}{5}, \infty\right)$ (ii) Decreasing in $\left[0, \frac{2}{5}\right]$
9	$\frac{dr}{dt} = -k$, a constant; so the radius decreases at a constant rate.
10	Local maximum value 12 (at $x = 0$); local minimum values -20 (at $x = -2$) and 7 (at $x = 1$)
11	f is increasing in $[-1, \infty)$
12	String is released at 1.6 m/s $(= 160$ cm/s)
13	Absolute maximum $= 56$ at $x = 5$; absolute minimum $= 24$ at $x = 1$
14	Maximum profit $= ₹\,513$ (at $x = 21$ units)

Level 3

15	Area is maximum when both legs equal $\frac{h}{\sqrt{2}}$ (isosceles); maximum area $= \frac{h^2}{4}$
16	Altitude $= \frac{4r}{3}$; maximum volume $= \frac{32\pi r^3}{81} = \frac{8}{27} \times \frac{4}{3}\pi r^3$
17	Width $= \frac{12}{6 - \sqrt{3}}$ m $(\approx 2.81$ m), height $= \frac{18 - 6\sqrt{3}}{6 - \sqrt{3}}$ m $(\approx 1.78$ m)
18	Strictly decreasing in $\left(0, \frac{\pi}{4}\right)$; strictly increasing in $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$
19	The water level rises at $\frac{5}{4\pi}$ m/h $(\approx 0.40$ m/h)
20	(i) $P(x) = -0.8x^2 + 30x - 150$ (ii) $x = 18.75$ (iii)(a) $P''(x) = -1.6 < 0$, so maximum profit (iii)(b) $x = 18$ bulbs, maximum profit $= ₹\,130.80$